Multi-Objective Optimization of the Period Traveling Salesman Problem via Genetic Algorithms

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1 INTRODUCTION

Market globalization and rapid technological advancements have led to an increased reliance on third-party service providers across logistics, distribution, and customer-facing industries. In this evolving landscape, characterized by growing service complexity and demand variability, the optimization of route planning has emerged as a critical determinant of operational efficiency and excellent service delivery. Effective route optimization directly contributes to reducing operational expenditures—such as fuel consumption, labor deployment, and vehicle wear—while simultaneously improving the reliability, responsiveness, and overall quality of service delivery [17].

Given the escalating complexity of service delivery models, the deployment of advanced route optimization techniques has become vital for sustaining operational resilience and achieving long-term strategic goals. Efficient management of routing not only minimizes costs but also enables organizations to remain competitive in increasingly dynamic markets [16]

The Period Travelling Salesman Problem (PTSP) focuses on determining the shortest possible route where a salesman moves from a starting point p visits a set of n customers exactly once before returning to p. However, many real-world applications—such as logistics, healthcare services, and emergency response [8] — demand more nuanced objectives, including the prioritization of early service to high-demand or critical customers. Hence, solving the PTSP from a multi-objective perspective is essential, as it aligns better with practical needs where total distance minimization must be balanced against timely service delivery.

In this study, an optimizer that simultaneously minimizes the total distance traveled and maximizes the early service of high-priority customers is designed. A Multi-Objective Genetic Algorithm (MOGA) framework is adopted to solve this multi-objective problem as it is well-suited to efficiently explore and preserve trade-offs between conflicting objectives, thus maintaining diverse solutions through an archive and providing a set of optimal solutions to select from based on priorities [1].

Author's address: XXXX XXXX #XXXXXX, xxx.xxx@xxx, xxxx, xxx, xx.

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2 BACKGROUND

The Travelling Salesman Problem (TSP) first formulated by Dantzig and Ramser [5], belongs to the class of NP-hard optimization problems. It involves determining the most cost-effective Hamiltonian cycle that visits each node in a weighted graph exactly once before returning to the starting point. The objective being to minimize the distance covered. Due to its combinatorial nature and computational complexity, the TSP serves as a benchmark for evaluating the performance of optimization algorithms and is applied across manufacturing, logitics and network design [3].

Over the decades, a variety of approaches have been developed to address the TSP and related routing challenges, including extensions such as the Vehicle Routing Problem (VRP) [14] and the Electric Vehicle Routing Problem (EVRP) [2], which emerged from advancements in transportation logistics and evolving operational requirements. The earliest approach to solving the TSP was brute force i.e trying every possible outcome, computing the distance and comparing solutions. In 1958, an algorithm was designed through the combined effort of Croes, Eastman, Bock [6] and named the "branch-and-bound" method by Little, Murty, Sweeney, and Karel [12]. It involved computing the bound on the best solution attainable traversing down the nodes connected to current node. These techniques were computationally infeasible for datasets involving a large amount of locations/cities/customers. Heuristic and metaheuristic algorithms such as Nearest Neighbor, the Christofides–Serdyukov algorithm, Ant colony optimization (ACO) and Lin–Kernighan heuristic [15] with the Lin-Kernighan being the best performing heuristic for the largest set problem [13]. Among metaheuristic techniques, Genetic Algorithms (GA) which were first developed in 1967 by Holland [11] proved to effectively solve the TSP reaching optimal solutions. GA simulates the natural selection process by preserving solutions which evolve through steps termed after biological processes - selection, crossover, and mutation. Solutions which are produced are called a population, with a single solution being an individual. The individuals are evaluated and maintained via a fitness operator; in the context of the TSP, the distance covered is evaluated; the shorter the distance, the fitter the individual and the less prone it is to being eliminated or replaced.

Real-world scenarios, however, involve multiple objectives; some of which might be conflicting; such as addressing varying casualties in an emergency situation or, in the case of the EVRP - minimizing cost while reducing the environmental impact. This leads to a multiobjective optimization problem, and solving this involves seeking a set of optimal solutions where improving one objective would lead to a minimal degradation of the other objective [10]. Broadly, there are two principal approaches to addressing multi-objective optimization problems. The first either involves combining the multiple objectives into a single objective and constructing a function based on that or maintaining a primary objective while the other objective is transformed into a constraint. Both methods importantly yield a single solution which differentiates it from the second, more widely favored approach which involves determining a Pareto optimal set — a collection of non-dominated solutions where no single objective can be improved without compromising at least one other. These solutions are then stored in an archive and presented for decision-making [1].

3 METHOD

3.1 Problem definition

There are n customers to be visited by a salesman. Each customer with coordinates x and y has a demand score d. This demand score d represents the level of priority for each customer. The salesman sets off from a starting point p, visits each customer exactly once, and returns to the starting point p.

3.2 Objectives

The two main objectives of this implementation are to *minimize the total distance traveled* by the salesman while *maximizing the weighted sum of customer demand satisfied*, with earlier visits to customers weighted more.

3.3 Design

The initial population, which serves as the parent individuals, are generated by randomly shuffling the order of the customer IDs. The individual contains a path, the total distance and a demand score. The path is the sequence representing a permutation of the customer IDs. It represents the order in which the customers are visited, with the trip starting at p - indexed 0 and ending at p. The distance is the sum of Euclidean distances between consecutive customers, and the demand score is the weighted sum of the demand, where customers visited earlier in the trip contribute more. It is mathematically computed as:

$$maximize \sum_{i=0}^{n} \left(1 - \frac{i}{n}\right) \cdot d_{p_i}$$
$$where\ d_{p_i} = \text{demand of customer}$$

The next step will involve parent individuals being selected using the tournament selection strategy - a process that involves selecting a subset of k sized individuals, then selecting the fittest individual from that subset [9]. The fittest are selected using an evaluator, with each individual evaluated based on the total distance.

$$minimize \sum_{i}^{n} \sum_{j}^{n} x_{ij} d_{ij}$$
$$where\ x_{ij} = \begin{Bmatrix} 1; salesman\ travels\ from\ city\ i\ to\ city\ j \\ 0; otherwise \end{Bmatrix}$$
$$d_{ij} = \text{distance from customer } i \text{ to } j$$

Subsequently, the selected parent individuals are recombined using the order crossover operation to reproduce child individuals. The order crossover (OX) operation picks a sub-segment starting from a crossover point from a parent and the other parent provides the rest of the sequence maintaining relative order and skipping already present values [7].

Parent 1	1	2	3	4	5	6	7	8	Parent 2	3	7	8	2	6	5	1	4
Child	-	3	4	5	6	-	-	-	Final child	7	3	4	5	6	8	2	1

Using the swap mutation strategy, the child individuals are then mutated, resulting in new individuals. The swap mutation strategy involves randomly selecting customers in an individual and swapping their positions [4]. The individuals are then inserted into an archive that maintains the Pareto front of non-dominated solutions. A solution is non-dominated if no other solution is better than it in both objectives (maximized customer demand satisfaction and minimized total distance).

4 EXPERIMENTAL SETUP

Algorithm 1 Multiobjective genetic algorithm

Input: Customer list with coordinates and demands
Parameters:
  Population size $P = 50$
  Number of generations $G = 100$
  Crossover rate $p_c = 0.8$
  Mutation rate $p_m = 0.2$
  Archive size $A = 100$
Output: Pareto optimal solution set $\mathcal{A}$
Initialize population $\mathcal{P}_0$ of P random individuals
for each individual $i \in \mathcal{P}_0$ **do**
  Evaluate $f_1(i)$: total distance
  Evaluate $f_2(i)$: demand satisfaction score
end for
Initialize archive $\mathcal{A} \leftarrow$ non-dominated individuals in $\mathcal{P}_0$
for generation $g = 1$ to G **do**
  Select parents
  Apply crossover with probability p_c
  Apply mutation with probability p_m
  Evaluate offspring on f_1 and f_2
  Update archive $\mathcal{A}$ with non-dominated solutions
end for
return $\mathcal{A}$: final Pareto front of solutions

The algorithm was implemented in Python and is publicly available at https://github.com/LuisChiej/PTSP-MOGA

5 RESULTS

The MOGA effectively evolved a diverse Pareto front, with a clear illustration of the trade-off between minimizing the total trip distance and maximizing customer demand satisfaction. The results revealed that satisfying high demand customers resulted in longer trip distances while shortening the trip distance resulted in forgoing early visitation of customers with high demand. While the size of the Pareto front varied across different instantiations, these fluctuations did not exhibit any consistent relationship with the population size, suggesting other factors were more influential in shaping the solution diversity. In particular, parameters such as mutation rate and the number of generations appeared to play a more decisive role in determining the size of the Pareto-optimal solutions, with a low mutation rate leading to more solutions and a high mutation rate, fewer. The number of generations is however the most decisive parameter in determining the size of the Pareto front with an increased number of generations resulting in more solutions. The resulting Pareto front plots display a smooth, well-distributed front, highlighting the algorithm's ability to maintain solution diversity across conflicting objectives.

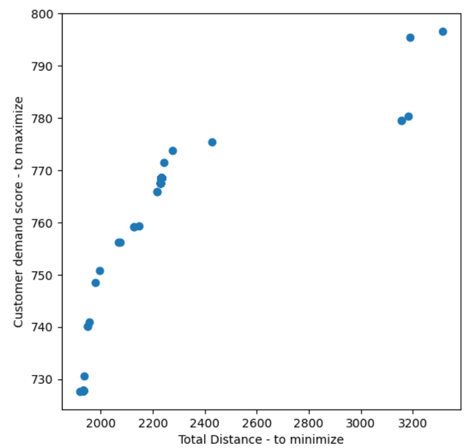

Fig. 1. Pareto front showing the trade-off between distance and demand.

6 CONCLUSION

In this study, a Multi-Objective Genetic Algorithm (MOGA) is proposed to address real-world routing scenarios where the goal of minimizing travel distance conflicts with maximizing customer satisfaction. The approach integrates order crossover, swap mutation, tournament selection, and Pareto-based solution archiving to effectively balance these competing objectives and present a diverse set of trade-off solutions. While the MOGA may not outperform metaheuristics such as Ant Colony Optimization (ACO) in pure distance minimization, it demonstrates superior performance in satisfying customer demand and achieves competitive results with reduced computational time.

Future work may incorporate additional real-world constraints, including time windows and salesman capacity limits, to enhance the realism of the model. Furthermore, extending the framework to accommodate multiple salesmen introduces new challenges. Random customer-to-salesman assignments can lead to imbalanced solutions, where some solutions exhibit low demand and short distances while others bear the opposite characteristics. This disparity can obscure global optimality unless the presentation of the pareto front is on a per-salesman basis.

Ultimately, the proposed approach underscores the value of multi-objective evolutionary methods in producing flexible, interpretable, and practically applicable solutions to the complex TSP where trade-offs are unavoidable.

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